

Rodrigo Alexander Fernandez Huarca

AI Engineer | Backend & Cloud-Native Systems

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Professional Summary

Software Engineer building applied AI systems and the backend infrastructure that runs them in production. Designs RAG pipelines end to end – document ingestion, embeddings, vector retrieval and context grounding – and ships them as containerized FastAPI services on Google Cloud Run. Comfortable across Python, C# (.NET) and TypeScript, and across GCP, Azure and AWS.

Technical Skills

- **AI & LLM:** RAG architectures, LangChain, prompt orchestration, Gemini API, Gemini Nano (on-device), OpenAI API, FAISS, sentence-transformers embeddings.
- **Machine Learning:** scikit-learn, HDBSCAN, K-Means, SciBERT, spaCy, TF-IDF, OpenCV, Pandas, NumPy.
- **Backend & APIs:** Python (FastAPI, Flask, Uvicorn), C# (.NET), TypeScript (NestJS, Node.js), Java (Spring Boot), REST, WebSockets.
- **Cloud & DevOps:** Google Cloud (Cloud Run), Azure, AWS (S3), Docker (multi-stage), Terraform, GitHub Actions, Grafana, Azure Monitor, Linux.
- **Databases:** PostgreSQL, PostGIS, FAISS (vector), MongoDB, SQLite.
- **Practices:** Layered/clean architecture, containerized environments, Git, Agile.
- **Languages:** Spanish (native), English (B2).

Work Experience

Software Engineer — Backend & AI Automation

May 2026 – Present

SimiAI

Remote

- Designed and led the proof of concept for a **Python agent engine** (FastAPI) with tool-orchestration logic, defining the skill registry and data flow for automated engineering design.
- Built the integration bridge that parses **AI-processed field data** – photogrammetry, GPS and NLP output – into standardized JSON payloads consumed downstream by the CAD automation pipeline.
- Programmed C# .NET logic that inserts dynamic blocks (poles, conduits, utility boxes) at precise georeferenced coordinates and orientations derived from routing parameters.
- Implemented asynchronous WPF/PaletteSet dockable panels (MVVM) so cloud-synced alerts and field media render without blocking the CAD UI thread.
- *Tech Stack:* C# .NET (AutoCAD API), Python (FastAPI), AWS (S3, SDK for .NET), Azure, Docker, PostgreSQL / PostGIS.

Jr. Software Engineer — Cloud & DevOps (CI/CD)

Nov 2025 – May 2026

Committed Organization

Remote

- Deployed and monitored backend inference services across **Azure and GCP**, building Grafana and Azure Monitor dashboards to track service health and support incident response.
- Automated cloud infrastructure provisioning with **Terraform** templates for reproducible, consistent environments.
- Configured Azure virtual networks, storage accounts and access policies; maintained CI/CD pipelines in GitHub Actions.
- *Tech Stack:* Azure, GCP, AWS, Terraform, Docker, Grafana, GitHub Actions, Bash, Linux.

Jr. Software Engineer — Full-Stack

Apr 2025 – Nov 2025

Committed Organization

Remote

- Built and maintained modular backend services in **Java and Spring Boot** for transactional data flows, containerized with Docker for parity between local and production environments.
- *Tech Stack:* Java, Spring Boot, React, PostgreSQL, Docker, Git.

Machine Learning Engineer — Research Project

Jun 2025 – Dec 2025

Universidad Nacional de San Agustín (UNSA)

Arequipa, Peru

- Led development and training of ML models (Python, scikit-learn) for **crop classification and phenological**

stage detection from satellite imagery, reaching +75% accuracy.

- Built Python/Pandas data pipelines with OpenCV preprocessing for feature extraction, **cutting manual analysis time by ~80%**.
- Containerized and deployed the inference services on Google Cloud Run.
- *Tech Stack*: Python, FastAPI, Pandas, OpenCV, scikit-learn, Google Earth Engine, Google Cloud Run, Docker.

AI Projects

Law Copilot — RAG over Peruvian legislation

Jan 2026

Lead AI & Cloud Engineer — Google AI Partner Catalyst

- Built a **specialized vector knowledge base** indexing the Political Constitution of Peru, the Civil Code, the Consumer Protection Code and the Law against Violence towards Women, so answers stay grounded in specific statutes.
- Implemented the retrieval layer with **FAISS** and sentence-transformers embeddings, using dynamic context injection to reduce hallucination on legal questions.
- Deployed the FastAPI backend to **Google Cloud Run** with multi-stage Docker builds to shrink the image and cut cold-start latency.
- *Tech Stack*: Python (FastAPI), LangChain, FAISS, Gemini API, OpenAI API, Next.js, Google Cloud Run, Docker.

Quechua Q&A — RAG for a low-resource language

2026

AI & Backend Engineer

- Built a question-answering system over Quechua grammar documents, returning answers with the source sections and page numbers used, so every response is traceable.
- Structured the FastAPI backend in **layered architecture** (API / services / repositories / core) with Pydantic models and async I/O.
- Set up GitHub Actions CI/CD with deployment configurations for both Azure and GCP.
- *Tech Stack*: Python (FastAPI), Gemini API, Pydantic, Next.js, TypeScript, GitHub Actions, Azure, GCP.

On-Device AI Study Assistant

Nov 2025

AI Engineer — Google Chrome Built-in AI Challenge 2025

- Integrated **Gemini Nano** inside Chrome to generate summaries, flashcards and five types of exercises **fully offline** – study documents never leave the user's device.
- Split the workload deliberately: local Nano for privacy-sensitive document processing, cloud Gemini via FastAPI and LangChain for learning roadmaps and generated games.
- *Tech Stack*: Next.js 14, TypeScript, Gemini Nano (Prompt API), Gemini API, Python (FastAPI), LangChain, PyPDF2.

Space Biology Knowledge Engine

Oct 2025

AI Engineer — NASA Space Apps Challenge 2025

- Clustered space-biology literature by meaning using **SciBERT embeddings and HDBSCAN**, with automatic cluster labeling (KeyBERT, TF-IDF) and an agent that extracts research evidence from each paper.
- Exposed the result as an interactive knowledge map backed by a FastAPI service.
- *Tech Stack*: Python (FastAPI), SciBERT, HDBSCAN, scikit-learn, spaCy, transformers, KeyBERT, React.

ONISAT CanSat 2026 — Telemetry & Imaging System

Dec 2025 – Present

Lead Software Engineer

- Owned the full data path for a CanSat mission: C++17 flight software on a Raspberry Pi Zero 2W, a 433 MHz LoRa downlink, and a Python ground station that reassembles fragmented images, recovering packets lost in flight with **OpenCV inpainting** instead of dropping frames.
- *Tech Stack*: C++17, Raspberry Pi, ESP32-S3, LoRa (SX1278), Python, OpenCV, WebSockets, Next.js, Three.js, Docker.

Education

Bachelor of Science in Systems Engineering

Expected 2027

Universidad Nacional de San Agustín (UNSA)

Arequipa, Peru